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389. On a Locus derived from Two Conics.

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I suppose in particular that the two conics are

$$x^2 + my^2 - 1 = 0, \quad mx^2 + y^2 - 1 = 0, \quad (1)$$

the equation of the quartic is

$$4(\lambda + 1)^2 m^2 (x^2 + my^2 - 1)(mx^2 + y^2 - 1) - \{(m^2 + m)(x^2 + y^2) - m^2 - 1\}^2 = 0, \quad (2)$$

or putting $\lambda = \frac{k}{m^2 + 1}$, this is

$$(\lambda + 1)^2 \{x^2 + y^2 - a\}^2 - (x^2 + my^2 - 1)(mx^2 + y^2 - 1) = 0. \quad (3)$$

To fix the ideas, suppose that m is positive and > 1 , so that each of the conics is an ellipse, the major semi-axis being $= 1$, and the minor semi-axis being $= \frac{1}{\sqrt{m}}$.

For any real value of k the coefficient λ is positive, and it may accordingly be assumed that λ is positive.

We have $\frac{1}{m(m+1)} < a < 1$, or the radius of the circle is intermediate between the semi-axes of the ellipses, hence the points of contact on each ellipse are real points.

Writing for shortness $a = \frac{m^2 + 1}{\lambda(m + 1)}$, the equation is

$$(x^2 + my^2 - 1)(mx^2 + y^2 - 1) - \lambda(x^2 + y^2 - a)^2 = 0. \quad (4)$$

For the points on the axis of x , we have

$$(x^2 - 1)(mx^2 - 1) - \lambda(x^2 - a)^2 = 0, \quad (5)$$

that is

$$(m - \lambda)x^4 + \{-(1 + m) + 2\lambda a\}x^2 + (1 - \lambda a^2) = 0, \quad (6)$$

and thence

$$(m - \lambda)x^2 = \frac{1}{2}(1 + m) - \lambda a \pm \frac{1}{2}\sqrt{(m - 1)^2 + 4\lambda(1 - a)(1 - ma)}, \quad (7)$$

or, substituting for a its value, this is

$$(m - \lambda)x^2 = \frac{1}{2}(m + 1) - \frac{\lambda(m^2 + 1)}{\lambda(m + 1)} \pm \frac{1}{2}\sqrt{(m - 1)^2 + 4\lambda(1 - a)(1 - ma)}. \quad (8)$$

Remarking that the values of λ : 0 , $\frac{1}{(m + 1)^2}$, $\frac{m}{(m + 1)^2}$, m , $\frac{1}{4}(m + 1)^2$ are in the order of increasing magnitude.

and considering successive values of λ ; first the value $\lambda = \frac{1}{(m+1)^2}$, we have

$$(m - \lambda)x^2 = \frac{1}{2}(m + 1) - \frac{1}{m+1} \pm \frac{1}{4}\sqrt{(m-1)(m - \frac{1}{m})}, \quad (9)$$

or observing that

$$(m + 1)^2(m + \frac{1}{m} - 2) = (m + 1)^2 - (m - 1)^2 = -(m - 1)(m^2 - 1) = (m - 1)(m - \frac{1}{m}), \quad (10)$$

this is

$$(m - \lambda)x^2 = 0, \quad (11)$$

or

$$x^2 = \frac{(m-1)(m - \frac{1}{m})}{(m+1)^2} \cdot \frac{(m+1)^2}{m(m+1)} = 0, \quad (12)$$

or, what is the same thing,

$$x^2 - \frac{(m-1)(m^3 + 2m^2 - 1)}{m(m+1)^2} = 0, \quad y^2 - \frac{(m-1)(m^3 + 2m^2 - 1)}{m(m+1)^2} = 0. \quad (13)$$

The next critical value is $\lambda = m$.

The curve here is

$$(x^2 + my^2 - 1)(mx^2 + y^2 - 1) - m(x^2 + y^2 - a)^2 = 0, \quad (14)$$

that is

$$m(x^2 + y^2)^2 + (1 + m^2)x^2y^2 - (m + 1)(x^2 + y^2) + 1 - m(x^2 + y^2)^2 + 2ma(x^2 + y^2) - ma^2 = 0, \quad (15)$$

that is

$$(m - 1)^2x^2y^2 + (2ma - m - 1)(x^2 + y^2) + 1 - ma^2 = 0, \quad (16)$$

or, substituting for a its value,

$$2ma - m - 1 = \frac{2m}{m+1} - (m+1) = \frac{-(m-1)^2}{m+1}, \quad (17)$$

$$1 - ma^2 = 1 - \frac{m(m^2 + 1)^2}{m^2(m+1)^2} = \frac{(m+1)^2(m^2 + 1) - m(m^2 + 1)^2}{m(m+1)^2} = \frac{(m^2 + 1)(-m^3 + m^2 + 2m + 1)}{m(m+1)^2}, \quad (18)$$

$$= -\frac{(m^2 + 1)(m^3 - m^2 - 2m - 1)}{m(m+1)^2}, \quad (19)$$

the equation is

$$(m - 1)^2x^2y^2 - \frac{(m-1)^2}{m+1}(x^2 + y^2) - \frac{(m^2 + 1)(m^3 - m^2 - 2m - 1)}{m(m+1)^2} = 0, \quad (20)$$

or, as this may also be written,

$$x^2y^2 - \frac{1}{m+1}(x^2 + y^2) - \frac{(m^2 + 1)(m^3 - m^2 - 2m - 1)}{m(m-1)^2(m+1)^2} = 0. \quad (21)$$

which has a pair of imaginary asymptotes parallel to the axis of x , and a like pair parallel to the axis of y , or what is the same thing, the curve has two isolated points at infinity, one on each axis.

The next critical value is $\lambda = \frac{1}{4}(m+1)^2$; the curve here reduces itself to the four lines

$$(x+y)^2 = \frac{m+1}{m}, \quad (x-y)^2 = \frac{1}{m}, \quad (21)$$

and it is to be observed that when λ exceeds this value, or say $\lambda > \frac{1}{4}(m+1)^2$, the curve has no real point on either axis; but when $\lambda = \infty$, the curve reduces itself to $(x^2 + y^2 - a)^2 = 0$, i.e. to the circle $x^2 + y^2 = a$ twice repeated, having in this special case real points on the two axes.

It is now easy to trace the curve for the different values of λ .

The curve lies in every case within the unshaded regions of the figure (except in the limiting cases after-mentioned); and it also touches the two ellipses and the four lines at the eight points k , at which points it also cuts the circle; but it does not cut or touch the four lines, the two ellipses, or the circle, except at the points k .

Considering λ as varying by successive steps from 0 to ∞ :

$\lambda = 0$, the curve is the two ellipses.

$\lambda < \frac{1}{(m+1)^2}$, the curve consists of two ovals, an exterior sinuous oval lying in the four regions a and the four regions b ; and an interior oval lying in the region c .

$\lambda = \frac{1}{(m+1)^2}$, there is still a sinuous oval as above, but the interior oval has dwindled to a conjugate point at the centre.

$\lambda = \frac{m}{(m+1)^2}$, there is still a sinuous oval as above, but the interior oval has dwindled to a conjugate point at the centre. For this critical value, the curve acquires the four lines as bitangents, touching each of them at two consecutive points k .

$\frac{m}{m+1} < \lambda < m$, there is no interior oval, but only a sinuous oval as above; which, as λ increases, approaches continually nearer to the four sides of the square.

$\lambda = m$, there is no change in the general form, but the curve has for this value of λ , two conjugate points, one on each axis at infinity.

$\lambda = \frac{1}{4}(m+1)^2$, the curve becomes the four lines.

$\lambda > \frac{1}{4}(m+1)^2$, the curve lies wholly in the four regions a and the four regions e , consisting thereof of four detached sinuous ovals. As λ deviates less from the value $\frac{1}{4}(m+1)^2$, each oval approaches more nearly to the infinite trilateral formed by the side and infinite line-portions which bound the regions d , e to which the oval belongs. And as λ departs from the limit $\frac{1}{4}(m+1)^2$, and approaches to ∞ , each sinuous oval approaches more nearly to the circular arc which separates the two regions d , e , which contains the sinuous oval. Finally, $\lambda = \infty$, the curve is the circle twice repeated.

390. Theorem relating to the Four Conics which touch the same Two Lines and pass through the same Four Points.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. VIII (1867), pp. 162–167.]

390. [Date Nov. 13, 1867.]

I.

The sides of the triangle formed by the given points meet one of the given lines in three points, say P , Q , R ; and on this same line we have four points of contact, say A_1 , A_2 , A_3 , A_4 ; any two pairs, say A_1 , A_2 ; A_3 , A_4 , form with a properly selected pair, say Q , R , out of the above-mentioned three points, an involution; and we have thus the three involutions

$$\{A_1, A_2; A_3, A_4; Q, R\}, \quad \{A_1, A_2; A_3, A_4; R, P\}, \quad \{A_1, A_2; A_3, A_4; P, Q\}. \quad (1)$$

To prove this, let $x = 0$, $y = 0$ be the equations of the given lines, and take for the equations of the sides of the triangle formed by the given points

$$bx + ay - ab = 0, \quad b'x + a'y - a'b' = 0, \quad b''x + a''y - a''b'' = 0; \quad (2)$$

the equation of any one of the four conics may be written

$$\frac{Lab}{bx + ay - ab} + \frac{L'a'b'}{b'x + a'y - a'b'} + \frac{L''a''b''}{b''x + a''y - a''b''} = Kxyz, \quad (3)$$

and if this touches the axis of x , say at the point $x = a$, then we must have

$$\frac{La}{x - a} + \frac{L'a'}{x - a'} + \frac{L''a''}{x - a''} = K(x - a). \quad (4)$$

or, assuming as we may do, $K = -(a' - a'')(a'' - a)(a - a')$, this gives

$$La = (a - a)^2(a' - a''), \quad L'a' = (a' - a)^2(a'' - a), \quad L''a'' = (a'' - a)^2(a - a'). \quad (5)$$

But in the same manner, if the conic touch the axis of y , say at the point $y = \beta$, we have

$$Lb = (b - \beta)^2(b' - b''), \quad L'b' = (b' - \beta)^2(b'' - b), \quad L''b'' = (b'' - \beta)^2(b - b'). \quad (6)$$

We have

$$\begin{aligned} b(a - \alpha)^2(a' - a'') : b'(a' - a)^2(a'' - a) : b''(a'' - a)^2(a - a') \\ = a(b - \beta)^2(b' - b'') : a'(b' - \beta)^2(b'' - b) : a''(b'' - \beta)^2(b - b'); \end{aligned} \quad (7)$$

and thence

$$\frac{(a - a)\sqrt{bb'}}{a - a} : \frac{(a' - a)\sqrt{b'b''}}{a' - a} : \frac{(a'' - a)\sqrt{b''b}}{a'' - a} = (b - \beta)(b' - b'') : (b' - \beta)(b'' - b) : (b'' - \beta)(b - b'). \quad (8)$$

Putting

$$P = ab(a' - a'')(b' - b''), \quad P' = a'b'(a'' - a)(b'' - b), \quad P'' = a''b''(a - a')(b - b'), \quad (9)$$

we have

$$(a - a)\sqrt{\frac{P}{a}} : (a' - a)\sqrt{\frac{P'}{a'}} : (a'' - a)\sqrt{\frac{P''}{a''}} = (b - \beta)(b' - b'') : (b' - \beta)(b'' - b) : (b'' - \beta)(b - b'), \quad (10)$$

and thence

$$\frac{(a - a)\sqrt{P}}{a} : \frac{(a' - a)\sqrt{P'}}{a'} : \frac{(a'' - a)\sqrt{P''}}{a''} = (b - \beta)(b' - b'') : (b' - \beta)(b'' - b) : (b'' - \beta)(b - b'). \quad (11)$$

which gives

$$(a - a)\sqrt{P} : (a' - a)\sqrt{P'} : (a'' - a)\sqrt{P''} = \frac{b - \beta}{\sqrt{aa'}}(b' - b'') : \frac{b' - \beta}{\sqrt{a'a''}}(b'' - b) : \frac{b'' - \beta}{\sqrt{a''a}}(b - b'). \quad (12)$$

and we have in like manner

$$(b - \beta)\sqrt{P} : (b' - \beta)\sqrt{P'} : (b'' - \beta)\sqrt{P''} = \frac{a - a}{\sqrt{bb'}}(a' - a'') : \frac{a' - a}{\sqrt{b'b''}}(a'' - a) : \frac{a'' - a}{\sqrt{b''b}}(a - a'); \quad (13)$$

but the first of these equations is alone required for the present purpose.

Putting for shortness

$$P = a^2\lambda, \quad P' = a'^2\lambda', \quad P'' = a''^2\lambda'', \quad (14)$$

the equation is

$$(a - a)\sqrt{(\lambda)} + (a' - a)\sqrt{(\lambda')} + (a'' - a)\sqrt{(\lambda'')} = 0, \quad (15)$$

and by attributing the signs + and – to the radicals, we have, corresponding to the four conics, the equations

$$(a - a_1)\sqrt{(\lambda)} + (a' - a_1)\sqrt{(\lambda')} + (a'' - a_1)\sqrt{(\lambda'')} = 0, \quad (14)$$

$$-(a - a_2)\sqrt{(\lambda)} + (a' - a_2)\sqrt{(\lambda')} + (a'' - a_2)\sqrt{(\lambda'')} = 0, \quad (15)$$

$$(a - a_3)\sqrt{(\lambda)} - (a' - a_3)\sqrt{(\lambda')} + (a'' - a_3)\sqrt{(\lambda'')} = 0, \quad (16)$$

$$(a - a_4)\sqrt{(\lambda)} + (a' - a_4)\sqrt{(\lambda')} - (a'' - a_4)\sqrt{(\lambda'')} = 0, \quad (17)$$

where a_1, a_2, a_3, a_4 are the values of a for the four conics respectively.

Eliminating a we obtain the system of three equations

$$(2a - a_1 - a_2)\sqrt{(\lambda)} + (a_2 - a_1)\sqrt{(\lambda')} + (a_2 - a_1)\sqrt{(\lambda'')} = 0, \quad (18)$$

$$(a_3 - a_4)\sqrt{(\lambda)} + (2a' - a_3 - a_4)\sqrt{(\lambda')} + (a_3 - a_4)\sqrt{(\lambda'')} = 0, \quad (19)$$

$$(a_1 + a_2 - a_3 - a_4)\sqrt{(\lambda)} + (a_1 + a_3 - a_2 - a_4)\sqrt{(\lambda')} + (a_1 + a_4 - a_2 - a_3)\sqrt{(\lambda'')} = 0, \quad (20)$$

and then eliminating the radicals we have

$$\begin{vmatrix} 2a - a_1 - a_2 & a_2 - a_1 & a_2 - a_1 \\ a_3 - a_4 & 2a' - a_3 - a_4 & a_3 - a_4 \\ a_1 + a_2 - a_3 - a_4 & a_1 + a_3 - a_2 - a_4 & a_1 + a_4 - a_2 - a_3 \end{vmatrix} = 0, \quad (21)$$

which is in fact

$$\begin{vmatrix} 1 & a + a' & aa' \\ 1 & a_1 + a_4 & a_1a_4 \\ 1 & a_2 + a_3 & a_2a_3 \end{vmatrix} = 0, \quad (22)$$

as may be verified by actual expansion; the transformation of the determinant is a peculiar one.

The foregoing result was originally obtained as follows, viz. writing for a moment

$$a\sqrt{(\lambda)} + a'\sqrt{(\lambda')} + a''\sqrt{(\lambda'')} = \theta, \quad \sqrt{(\lambda)} + \sqrt{(\lambda')} + \sqrt{(\lambda'')} = \Phi, \quad (20)$$

the four equations are

$$\theta - a_1\Phi = 0, \quad \theta - a_2\Phi = 2(a - a_2)\sqrt{(\lambda)}, \quad \theta - a_3\Phi = 2(a' - a_3)\sqrt{(\lambda')}, \quad \theta - a_4\Phi = 2(a'' - a_4)\sqrt{(\lambda'')}. \quad (21)$$

These give

$$(a_1 - a_2)\theta = 2(a - a_2)\sqrt{(\lambda)}, \quad (22)$$

$$(a_1 - a_3)\theta = 2(a' - a_3)\sqrt{(\lambda')}, \quad (23)$$

$$(a_1 - a_4)\theta = 2(a'' - a_4)\sqrt{(\lambda'')}. \quad (24)$$

From the last equation we have

$$\begin{aligned} (a_1 - a_4)\Phi &= 2\{\theta - a_4\sqrt{(\lambda)} - a'\sqrt{(\lambda')}\} - 2a_4\{\Phi - \sqrt{(\lambda)} - \sqrt{(\lambda')}\} \\ &= 2(a - a_4)\Phi - 2(a - a_4)\sqrt{(\lambda)} - 2(a' - a_4)\sqrt{(\lambda')}; \end{aligned} \quad (25)$$

that is

$$(a_1 - a_4)\Phi - 2(a - a_4)\sqrt{(\lambda)} - 2(a' - a_4)\sqrt{(\lambda')} = 0; \quad (26)$$

or substituting for $\sqrt{(\lambda)}$, $\sqrt{(\lambda')}$ their values in terms of Φ , we find

$$\frac{(a - a_4)(a_1 - a_2)}{a - a_2} + \frac{(a' - a_4)(a_1 - a_3)}{a' - a_3} = a_1 - a_4, \quad (27)$$

which may be written

$$\frac{(a - a_4)(a - a_1)}{a - a_2} + \frac{(a' - a_4)(a' - a_1)}{a' - a_3} = 0; \quad (28)$$

that is

$$(a_2 - a_1) \left(1 + \frac{a_2 - a_1}{a - a_2}\right) (a - a_4) + (a_3 - a_1) \left(1 + \frac{a_3 - a_1}{a' - a_3}\right) (a' - a_4) = 0; \quad (29)$$

or again

$$(a - a_2)(a - a_4) + (a' - a_3)(a' - a_4) + (a_2 - a_1)(a - a_4) \frac{a - a_2}{a - a_2} + (a_3 - a_1)(a' - a_4) \frac{a' - a_3}{a' - a_3} = 0; \quad (30)$$

that is

$$(a - a_2)(a - a_4) + (a' - a_3)(a' - a_4) + (a_2 - a_1)(a - a_4) + (a_3 - a_1)(a' - a_4) = 0; \quad (31)$$

or finally

$$(a_2 - a_1)(a - a_4)(a' - a_3) + (a_3 - a_4)(a - a_1)(a' - a_2) = 0, \quad (32)$$

which is a known form of the relation

$$\begin{vmatrix} 1 & a + a' & aa' \\ 1 & a_1 + a_4 & a_1a_4 \\ 1 & a_2 + a_3 & a_2a_3 \end{vmatrix} = 0, \quad (33)$$

which gives the involution of the quantities $a, a'; a_1, a_4; a_2, a_3$.

We have in like manner

$$\begin{vmatrix} 1 & a' + a'' & a'a'' \\ 1 & a_1 + a_3 & a_1a_3 \\ 1 & a_2 + a_4 & a_2a_4 \end{vmatrix} = 0, \quad \begin{vmatrix} 1 & a'' + a & a''a \\ 1 & a_1 + a_2 & a_1a_2 \\ 1 & a_3 + a_4 & a_3a_4 \end{vmatrix} = 0, \quad (34)$$

which give the involutions of the systems $a', a''; a_1, a_3; a_2, a_4$ and $a'', a; a_1, a_2; a_3, a_4$ respectively.

It may be remarked that the equation of the conic passing through the three points and touching the axis of x in the point $x = a$ is

$$\frac{(a-a')^2(a''-a'')b}{bx+ay-ab} + \frac{(a'-a)^2(a''-a)b'}{b'x+a'y-a'b'} + \frac{(a''-a)^2(a-a')b''}{b''x+a''y-a''b''} = 0, \quad (34)$$

and when this meets the axis of y we have

$$\frac{(a-a')^2(a''-a'')}{y-b} + \frac{(a'-a)^2(a''-a)}{y-b'} + \frac{(a''-a)^2(a-a')}{y-b''} = 0. \quad (35)$$

Hence, if this touches the axis of y in the point $y = \beta$, the left-hand side must be

$$R \frac{(a-a')^2(a''-a'') + (a'-a)^2(a''-a) + (a''-a)^2(a-a')}{(y-b)(y-b')(y-b'')} (y-\beta)^2 \quad (36)$$

and equating the coefficients of $\frac{1}{y}$, we have

$$\begin{aligned} & -(a-a')^2(a''-a'') + (a'-a)^2(a''-a) + (a''-a)^2(a-a') \\ & = [-(a-a')^2(a''-a'') + (a'-a)^2(a''-a) + (a''-a)^2(a-a')](b+b'+b''-2\beta), \end{aligned} \quad (37)$$

or what is the same thing,

$$\begin{aligned} & \frac{(a-a')^2(a''-a'')}{a} + \frac{(a'-a)^2(a''-a)}{a'} + \frac{(a''-a)^2(a-a')}{a''} \\ & \times \{-(a-2\beta)(a'-a'') + (a'-2\beta)(a''-a) + (a''-2\beta)(a-a')\} = 0, \end{aligned} \quad (38)$$

which gives β in terms of a , that is $\beta_1, \beta_2, \beta_3, \beta_4$ in terms of a_1, a_2, a_3, a_4 respectively.

Cambridge, 30 November, 1863.

391. Solution of a Problem of Elimination.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. VIII (1867), pp. 183–185.]

391. [Date Nov. 13, 1867.]

It is required to eliminate x, y from the equations

$$\begin{array}{ccccc} x^4, & x^3y, & x^2y^2, & xy^3, & y^4 \\ a, & b, & c, & d, & e \\ a', & b', & c', & d', & e' \\ a'', & b'', & c'', & d'', & e'' \end{array} = 0. \quad (1)$$

This system may be written

$$x^4 = \Sigma \lambda a, \quad x^3y = \Sigma \lambda b, \quad x^2y^2 = \Sigma \lambda c, \quad xy^3 = \Sigma \lambda d, \quad y^4 = \Sigma \lambda e; \quad (2)$$

or putting

$$y = kx, \quad (3)$$

we have

$$\Sigma \lambda(a + kb) = 0, \quad \Sigma \lambda(b + kc) = 0, \quad \Sigma \lambda(c + kd) = 0, \quad \Sigma \lambda(d + ke) = 0; \quad (4)$$

or, what is the same thing,

$$\lambda(a + kb) + \lambda'(a' + kb') + \lambda''(a'' + kb'') = 0, \quad (5)$$

$$\lambda(b + kc) + \lambda'(b' + kc') + \lambda''(b'' + kc'') = 0, \quad (6)$$

$$\lambda(c + kd) + \lambda'(c' + kd') + \lambda''(c'' + kd'') = 0, \quad (7)$$

$$\lambda(d + ke) + \lambda'(d' + ke') + \lambda''(d'' + ke'') = 0; \quad (8)$$

and representing the columns

$$\begin{array}{cc|cc|cc} a & b, & a' & b', & a'' & b'' \\ b & c, & b' & c', & b'' & c'' \\ c & d, & c' & d', & c'' & d'' \\ d & e, & d' & e', & d'' & e'' \end{array} \quad (9)$$

by 1, 2, 3, 4, 5, 6; each equation is of the type

$$\lambda(1 + k \cdot 2) + \lambda'(3 + k \cdot 4) + \lambda''(5 + k \cdot 6) = 0. \quad (10)$$

Multiplying the several equations by the minors of 135, each with its proper sign, and adding, the terms independent of k disappear, the equation divides by k , and we find

$$\lambda \cdot 2135 + \lambda' \cdot 4135 + \lambda'' \cdot 6135 = 0; \quad (11)$$

operating in a similar manner with the minors of 246, the terms in k disappear, and we find

$$\lambda \cdot 1246 + \lambda' \cdot 3246 + \lambda'' \cdot 5246 = 0; \quad (12)$$

again, operating with the minors of $(146 + 236 + 245 + k \cdot 246)$, we find

$$\begin{aligned} & \lambda\{1236 + 1245 + k(2146 + 1246)\} \\ & + \lambda'\{3146 + 3245 + k(4236 + 3246)\} \\ & + \lambda''\{5146 + 5236 + k(6245 + 5246)\} = 0, \end{aligned} \quad (13)$$

where the terms in k disappear, and this is

$$\lambda(1236 + 1245) + \lambda'(3146 + 3245) + \lambda''(5146 + 5236) = 0. \quad (14)$$

We have thus three linear equations, which written in a slightly different form are

$$\lambda \cdot 1235 + \lambda' \cdot 3451 + \lambda'' \cdot 5613 = 0, \quad (15)$$

$$\lambda(1236 + 1245) + \lambda'(3452 + 3461) + \lambda''(5614 + 5623) = 0, \quad (16)$$

$$\lambda \cdot 1246 + \lambda' \cdot 3462 + \lambda'' \cdot 5624 = 0, \quad (17)$$

and thence eliminating $\lambda, \lambda', \lambda''$, we have

$$\begin{vmatrix} 1235, & 3451, & 5613 \\ 1236 + 1245, & 3452 + 3461, & 5614 + 5623 \\ 1246, & 3462, & 5624 \end{vmatrix} = 0, \quad (18)$$

which is the required result.

It may be remarked that the second and third column are obtained from the first by operating on it with $\Delta, \frac{1}{2}\Delta^2$, if $\Delta = 2\delta_1 + 4\delta_2 + 6\delta_3$.

Or say the result is

$$(1, \Delta, \frac{1}{2}\Delta^2) \begin{vmatrix} 1235 \\ 3451 \\ 5613 \end{vmatrix} = 0. \quad (19)$$

In like manner for the system

$$\begin{array}{cccccc}
 x^5, & x^4y, & x^3y^2, & x^2y^3, & xy^4, & y^5 \\
 a, & b, & c, & d, & e, & f \\
 a', & b', & c', & d', & e', & f' \\
 a'', & b'', & c'', & d'', & e'', & f'' \\
 a''', & b''', & c''', & d''', & e''', & f'''
 \end{array} \tag{18}$$

if the columns are

$$\begin{array}{cc|cc|cc|cc}
 a & b, & a' & b', & a'' & b'', & a''' & b''' \\
 b & c, & b' & c', & b'' & c'', & b''' & c''' \\
 c & d, & c' & d', & c'' & d'', & c''' & d''' \\
 d & e, & d' & e', & d'' & e'', & d''' & e''' \\
 e & f, & e' & f', & e'' & f'', & e''' & f'''
 \end{array} \tag{19}$$

$= 1, 2, 3, 4, 5, 6, 7, 8$; then the result is

$$(1, \Delta, \tfrac{1}{2}\Delta^2, \tfrac{1}{6}\Delta^3) \begin{vmatrix} 12357 \\ 34571 \\ 56713 \\ 78135 \end{vmatrix} = 0, \tag{20}$$

where

$$\Delta = 2\delta_1 + 4\delta_3 + 6\delta_5 + 8\delta_7. \tag{21}$$

392. On the Conics which pass through Two Given Points and touch Two Given Lines.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. VIII (1867), pp. 211–219.]

392.

Let $x = 0$, $y = 0$ be the equations of the given lines; $z = 0$ the equation of the line joining the given points.

We may, to fix the ideas, imagine the implicit constants so determined that $x + y + z = 0$ shall be the equation of the line infinity.

Take $x - my = 0$, $x - ny = 0$ as the equations of the lines which by their intersection with $z = 0$ determine the given points.

The equation of the conic is

$$(x - my)(x - ny) + 2\{x + y\sqrt{(mn)}\}\sqrt{(yz)} = x + y\sqrt{(mn)} + \gamma z, \quad (1)$$

or, what is the same thing,

$$(x - my)(x - ny) + 2\{x + y\sqrt{(mn)}\}\gamma z + \gamma^2 z^2 = 0, \quad (2)$$

so that there are two distinct series of conics according as $\sqrt{(mn)}$ is taken with the positive or the negative sign.

The equation of the chord of contact is

$$x + y\sqrt{(mn)} + \gamma z = 0, \quad (3)$$

which meets $z = 0$ in the point $[x + y\sqrt{(mn)} = 0, z = 0]$, that is in one of the centres of the involution formed by the lines $(x = 0, y = 0)$, $(x - my = 0, x - ny = 0)$.

It is to be observed that the conic is only real when mn is positive, that is (the lines and points being each real) the two points must be situate in the same region or in opposite regions of the four regions formed by the two lines: there are however other real cases; e.g. if the lines $x = 0$, $y = 0$ are real, but the quantities m , n are conjugate imaginaries; included in this we have the circles which touch two real lines.

To fix the ideas I take m and n each positive and $mn > 1$; also I attend first to the series where $\sqrt{(mn)}$ is taken positively.

At the points where the conic meets infinity, we have

$$\{\sqrt{(m)} + \sqrt{(n)}\}\sqrt{(xy)} = x + y\sqrt{(mn)} - \gamma(x + y), \quad (4)$$

which gives two coincident points, that is the conic is a parabola, if

$$(1 - \gamma)\{\sqrt{(mn)} - \gamma\} = \frac{1}{4}\{\sqrt{(m)} + \sqrt{(n)}\}^2, \quad (5)$$

that is

$$\gamma = \frac{1}{2}\{1 + \sqrt{(mn)} \pm \sqrt{(1 + m)(1 + n)}\}, \quad (6)$$

where it is to be noticed that

$$\gamma = \frac{1}{2}[1 + \sqrt{(mn)} + \sqrt{\{(1 + m)(1 + n)\}}] \quad (7)$$

is a positive quantity greater than $\sqrt{(mn)}$, say $\gamma = p$;

$$\gamma = \frac{1}{2}[1 + \sqrt{(mn)} - \sqrt{\{(1 + m)(1 + n)\}}] \quad (8)$$

is a negative quantity, say $\gamma = -q$, q being positive.

The order of the lines is as shown in fig. 1, see plate facing p. 52.

$\gamma = -\infty$ to $\gamma = -q$, curve is ellipse;

$\gamma = -q$, parabola P_1 ;

$\gamma = -q$ to $\gamma = p$, curve is hyperbola;

$\gamma = p$, parabola P_2 ;

$\gamma = p$ to $\gamma = \infty$, ellipse.

Resuming the equation

$$(x - my)(x - ny) + 2\{x + y\sqrt{(mn)}\}yz + \gamma^2 z^2 = 0, \quad (9)$$

the coefficients are

$$(a, b, c, f, g, h) = (1, mn, \gamma^2, \gamma\sqrt{(mn)}, \gamma, -\frac{1}{2}(m + n)), \quad (10)$$

and thence the inverse coefficients are

$$\begin{aligned} (A, B, C, F, G, H) = & (0, 0, -\frac{1}{4}(m - n)^2, -\frac{1}{4}\gamma\{\sqrt{(m)} + \sqrt{(n)}\}^2, \\ & -\frac{1}{4}\gamma\sqrt{(mn)}\{\sqrt{(m)} + \sqrt{(n)}\}^2, \frac{1}{4}\gamma^2\{\sqrt{(m)} + \sqrt{(n)}\}^2). \end{aligned} \quad (11)$$

$$K = -\frac{1}{4}\gamma^2\{\sqrt{(m)} + \sqrt{(n)}\}^2,$$

or, omitting a factor, the inverse coefficients are

$$(A, B, C, F, G, H) = (0, 0, -\gamma\{\sqrt{(m)} - \sqrt{(n)}\}^2, 1, \sqrt{(mn)}, -\gamma). \quad (12)$$

Considering the line $\lambda x + \mu y + \nu z = 0$, the coordinates of the pole of this line are

$$X : Y : Z = -\gamma\mu + \sqrt{(mn)}\nu : -\gamma\lambda + \nu : \sqrt{(mn)}\lambda + \mu + \frac{1}{2}\{\sqrt{(m)} - \sqrt{(n)}\}^2\nu, \quad (12)$$

or (what is the same thing) introducing the arbitrary coefficient k , we have

$$kx + y\mu - \sqrt{(mn)} = 0, \quad (13)$$

$$ky + \gamma\lambda - \mu = 0, \quad (14)$$

$$kz - \lambda\sqrt{(mn)} - \mu - \frac{1}{2}\{\sqrt{(m)} - \sqrt{(n)}\}^2\nu = 0; \quad (15)$$

the first two equations give

$$k : \gamma : -1 = \nu\{\mu - \lambda\sqrt{(mn)}\} : \nu\{\gamma\sqrt{(mn)} - \lambda\} : \lambda x - \mu y, \quad (16)$$

that is

$$k = \frac{-\nu\{\mu - \lambda\sqrt{(mn)}\}}{\lambda x - \mu y}, \quad \gamma = \frac{-\nu\{\gamma\sqrt{(mn)} - \lambda\}}{\lambda x - \mu y}, \quad (17)$$

or, substituting this value of γ in the third equation,

$$\begin{aligned} \frac{\nu\{\mu - \lambda\sqrt{(mn)}\}(x - y\sqrt{(mn)})}{\lambda x - \mu y} + \frac{\{\mu + \lambda\sqrt{(mn)}\}(\lambda x - \mu y)}{\lambda x - \mu y} \\ + \frac{\frac{1}{2}\{\sqrt{(m)} - \sqrt{(n)}\}^2\nu(x - y\sqrt{(mn)})}{\lambda x - \mu y} = 0, \end{aligned} \quad (18)$$

that is

$$\begin{aligned} (\lambda x - \mu y)^2 \cdot \frac{1}{2}\{\sqrt{(m)} - \sqrt{(n)}\}^2 + \{x - y\sqrt{(mn)}\}(\lambda x - \mu y)\{\mu + \lambda\sqrt{(mn)}\} \\ + z\{x - y\sqrt{(mn)}\}\nu\{\mu - \lambda\sqrt{(mn)}\} = 0, \end{aligned} \quad (19)$$

which is the equation of the curve, the locus of the pole of the line $\lambda x + \mu y + \nu z = 0$ in regard to the conic $(x - my)(x - ny) + 2\{x + y\sqrt{(mn)}\}yz + \gamma^2 z^2 = 0$.

In particular, if $\lambda = \mu = \nu = 1$, then for the coordinates of the centre of the conic, we have

$$X : Y : Z = -\gamma + \sqrt{(mn)} : -\gamma + 1 : \sqrt{(mn)} + 1 + \frac{1}{2}\{\sqrt{(m)} - \sqrt{(n)}\}^2; \quad (18)$$

and for the locus of the centre,

$$\begin{aligned} (X - Y)^2 \cdot \frac{1}{2}\{\sqrt{(m)} - \sqrt{(n)}\}^2 + (x - y)\{x - y\sqrt{(mn)}\}\{1 + \sqrt{(mn)}\} \\ + z\{x - y\sqrt{(mn)}\}\{1 - \sqrt{(mn)}\} = 0, \end{aligned} \quad (19)$$

so that the locus is a conic, and it is obvious that this conic is a hyperbola.

Putting for greater simplicity

$$x - y = X, \quad x - y\sqrt{(mn)} = Y, \quad z = Z, \quad (20)$$

the equation of the curve of centres is

$$Z^2 \cdot \frac{1}{2}\{\sqrt{(m)} - \sqrt{(n)}\}^2 + XY\{1 + \sqrt{(mn)}\} + YZ\{1 - \sqrt{(mn)}\} = 0, \quad (21)$$

or, writing this under the form

$$Y[Z\{1 + \sqrt{(mn)}\} + Z\{1 - \sqrt{(mn)}\}] + \frac{1}{2}\{\sqrt{(m)} - \sqrt{(n)}\}^2 Z^2 = 0, \quad (22)$$

the equation is

$$XY + ZQ = 0, \quad (22)$$

where

$$X = x - y, \quad (23)$$

$$Y = x - y\sqrt{(mn)}, \quad (24)$$

$$Q = \frac{2}{\{\sqrt{(m)} - \sqrt{(n)}\}^2} [Z\{1 + \sqrt{(mn)}\}(x - y) + \{1 - \sqrt{(mn)}\}z]; \quad (25)$$

these values give

$$x - y = X, \quad (26)$$

$$x - y\sqrt{(mn)} = Y, \quad (27)$$

$$\{1 - \sqrt{(mn)}\}z = \frac{1}{2}\{\sqrt{(m)} - \sqrt{(n)}\}^2 Q + 2\{1 + \sqrt{(mn)}\}X, \quad (28)$$

or, what is the same thing,

$$\{1 - \sqrt{(mn)}\}x = -\sqrt{(mn)}X + Y, \quad (29)$$

$$\{1 - \sqrt{(mn)}\}y = -X + Y, \quad (30)$$

$$\{1 - \sqrt{(mn)}\}z = 2\{1 + \sqrt{(mn)}\}X + \frac{1}{2}\{\sqrt{(m)} - \sqrt{(n)}\}^2 Q. \quad (31)$$

whence also

$$\{1 - \sqrt{(mn)}\}(x + y + z) = \{1 + \sqrt{(mn)}\}Z + 2Y + \{\sqrt{(m)} - \sqrt{(n)}\}^2 Q, \quad (32)$$

or the equation of the line infinity is

$$\{1 + \sqrt{(mn)}\}X + 2Y + \{\sqrt{(m)} - \sqrt{(n)}\}^2 Q = 0, \quad (33)$$

a formula which may be applied to finding the asymptotes and thence the centre of the conic $YQ + X^2 = 0$.

In fact we have identically

$$\{2kx + 2ky - (2k + 1)z\}^2 - (1 + 4k)(2kx - z)^2 = 4k^2(x + y + z)^2 - 4k(1 + 4k)(kx^2 + yz), \quad (34)$$

that is

$$-4k(1 + 4k)(kx^2 + yz) = \{2kx + 2ky - (2k + 1)z\}^2 - (1 + 4k)(2kx - z)^2 - 4k^2(x + y + z)^2, \quad (35)$$

which, if $x + y + z = 0$ is the equation of the line infinity, puts in evidence the asymptotes of the conic $kx^2 + yz = 0$.

Hence writing αx , βy , γz in the place of x , y , z respectively, and $k = k'$, that is, $k = \frac{\beta\gamma}{\alpha^2}k'$, we have

$$\begin{aligned} -4\beta\gamma k'(\alpha^2 + 4\beta\gamma k')(k'x^2 + yz) &= \{2\alpha\gamma k'x + 2\beta^2 k'y - (2\beta\gamma k' + \alpha^2)z\}^2 \\ &\quad - (\alpha^2 + 4\beta\gamma k')(2\beta\gamma k'x - \alpha z)^2 - 4\beta^2\gamma^2 k'^2(\alpha x + \beta y + \gamma z)^2. \end{aligned} \quad (31)$$

Now writing X , Y , Q in the place of x , y , z ; $k' = 1$, and $\alpha = \{1 + \sqrt{(mn)}\}$, $\beta = 2$, $\gamma = \{\sqrt{(m)} - \sqrt{(n)}\}^2$, we have

$$\begin{aligned} &-16[\{1 + \sqrt{(mn)}\}^2 + 8\{\sqrt{(m)} - \sqrt{(n)}\}^2](YQ + X^2) \\ &= [4\{1 + \sqrt{(mn)}\}X + 8Y - (4\{\sqrt{(m)} - \sqrt{(n)}\}^2 + \{1 + \sqrt{(mn)}\}^2)Q]^2 \\ &\quad - [\{1 + \sqrt{(mn)}\}^2 + 8\{\sqrt{(m)} - \sqrt{(n)}\}^2][4X - \{1 + \sqrt{(mn)}\}Q]^2 \\ &\quad - 16[\{1 + \sqrt{(mn)}\}X + 2Y + \{\sqrt{(m)} - \sqrt{(n)}\}^2Q]^2, \end{aligned} \quad (32)$$

and the asymptotes are

$$\begin{aligned} &4\{1 + \sqrt{(mn)}\}X + 8Y - [4\{\sqrt{(m)} - \sqrt{(n)}\}^2 + \{1 + \sqrt{(mn)}\}^2]Q \\ &= \pm \sqrt{\{1 + \sqrt{(mn)}\}^2 + 8\{\sqrt{(m)} - \sqrt{(n)}\}^2} [4X - \{1 + \sqrt{(mn)}\}Q]. \end{aligned} \quad (33)$$

At the centre

$$4(1 + \sqrt{(mn)})X + 8Y - [4\{\sqrt{(m)} - \sqrt{(n)}\}^2 + \{1 + \sqrt{(mn)}\}^2]Q = 0, \quad (34)$$

$$4X - \{1 + \sqrt{(mn)}\}Q = 0, \quad (35)$$

but the first equation is

$$\{1 + \sqrt{(mn)}\}[4X - Q\{1 + \sqrt{(mn)}\}] + 8Y - 4\{\sqrt{(m)} - \sqrt{(n)}\}^2Q = 0, \quad (36)$$

so that we have

$$4X = \{1 + \sqrt{(mn)}\}Q, \quad (37)$$

$$2Y = \{\sqrt{(m)} - \sqrt{(n)}\}^2Q, \quad (38)$$

the first of these is

$$2\{\sqrt{(m)} - \sqrt{(n)}\}^2(x - y) - \{1 + \sqrt{(mn)}\}^2(x - y) - (1 - mn)z = 0, \quad (39)$$

and the two together give

$$2Z\{\sqrt{(m)} - \sqrt{(n)}\}^2 - \{1 + \sqrt{(mn)}\}Y = 0, \quad (40)$$

so that we have

$$2\{\sqrt{(m)} - \sqrt{(n)}\}^2(x - y) - \{1 + \sqrt{(mn)}\}(x - y\sqrt{(mn)}) - (1 - mn)z = 0. \quad (41)$$

393. On the Conics which touch Three Given Lines, &c.

[From the *Proceedings of the London Mathematical Society*, vol. IV (1871), pp. 114–115.]

393.

The problem of determining the conics which touch three given lines and satisfy some other condition is considered. When the three lines are given, the conic is determined by three tangential conditions. If additional conditions are given, such as passing through a point or touching a fourth line, the conic is completely determined.

The analytical treatment of this problem leads to interesting equations. Let the three given lines be

$$\lambda x + \mu y + \nu z = 0, \quad \lambda' x + \mu' y + \nu' z = 0, \quad \lambda'' x + \mu'' y + \nu'' z = 0. \quad (1)$$

The condition that a conic touches these three lines can be expressed by means of the tangential equation of the conic.

[Plate I facing page 52: Figure illustrating the conics touching three given lines.]

394. On a Locus in relation to the Triangle.

[From the *Proceedings of the London Mathematical Society*, vol. IV (1871), pp. 115–121.]

394.

The problem of finding the locus of a point related in a given manner to a triangle leads to many interesting curves.

Let the triangle be ABC , and let the point P be such that its distances from the sides have a given relation. The locus of P is in general a curve of higher order.

Consider the case where the point is the intersection of lines drawn from the vertices with given directions. The locus can be represented by the equation

$$\begin{vmatrix} A & H & G \\ H & B & F \\ G & F & C \end{vmatrix} = 0. \tag{1}$$

Consider the case where the point is such that the perpendiculars from it on the sides of the triangle have their feet in a line.

Let the sides of the triangle be $x = 0$, $y = 0$, $z = 0$, and let the Absolute (the conic in regard to which the perpendiculars are drawn) be

$$(a, b, c, f, g, h \mid x, y, z)^2 = 0. \quad (2)$$

The condition that the feet of the perpendiculars lie in a line is that the point (x, y, z) satisfies a certain cubic equation.

that is, the locus is a cubic curve passing through the vertices of the triangle.

The condition

$$\frac{abc}{af^2} \cdot \frac{1}{bg^2} \cdot \frac{1}{ch^2} = \frac{1}{fgh} \quad (3)$$

is the condition in order that the function

$$\frac{\xi^2}{a} + \frac{\eta^2}{b} + \frac{\zeta^2}{c} \quad (4)$$

may break up into linear factors; the function in question is

$$\frac{bc}{f^2} + \frac{ca}{g^2} + \frac{ab}{h^2}, \quad (5)$$

which is

$$\left(\frac{bc}{f}, \frac{ca}{g}, \frac{ab}{h} \mid x, y, z \right)^2 = (a, b, c, f, g, h \mid x, y, z)^2 + 2 \left(\frac{f}{a}yz + \frac{g}{b}zx + \frac{h}{c}xy \right), \quad (6)$$

so that the condition is, that the conic

$$(a, b, c, f, g, h \mid x, y, z)^2 + 2 \left(\frac{f}{a}yz + \frac{g}{b}zx + \frac{h}{c}xy \right) = 0, \quad (7)$$

(which is a certain conic passing through the intersections of the Absolute $(a, b, c, f, g, h \mid x, y, z)^2 = 0$, and of the locus conic $\frac{f}{a}yz + \frac{g}{b}zx + \frac{h}{c}xy = 0$) shall be a pair of lines.

Writing the equation of the conic in question under the form

$$\left(\frac{bc}{f}, \frac{ca}{g}, \frac{ab}{h} \mid x, y, z \right)^2, \quad (8)$$

the inverse coefficients A', B', C', F', G', H' of this conic, are

$$\frac{A'bc}{f^2} = \frac{B'ca}{g^2} = \frac{C'ab}{h^2} = \frac{abc}{fgh}, \quad (9)$$

so that we have $F' : G' : H' = F : G : H$.

Hence, if in regard to this new conic we form the reciprocal of the triangle ($x = 0, y = 0, z = 0$), and join the corresponding angles of the two triangles, the joining lines meet in a point which is the same point as is obtained by the like process from the triangle and its reciprocal in regard to the Absolute.

But I do not further pursue this part of the theory.

It is to be noticed that the conic

$$\frac{A}{f}yz + \frac{B}{g}zx + \frac{C}{h}xy = 0, \quad (10)$$

contains the angles of the reciprocal triangle, and is thus in fact the conic in which are situate the angles of the two triangles.

For the coordinates of one of the angles of the reciprocal triangle are (A, H, G) ; we should therefore have

$$\frac{A}{f}HG + \frac{B}{g}GA + \frac{C}{h}AB = 0, \quad (11)$$

which is $\frac{1}{f}(GH \cdot gh + BG \cdot hf + GH \cdot fg) = 0$, or attending only to the second factor and writing $GH = Kf + AF$, the condition is

$$Kfgh + AFgh + BGhf + CHfg = 0, \quad (12)$$

or substituting for K, A, B, C, F, G, H their values and reducing, this is satisfied: hence the three angles of the reciprocal triangle lie on the conic in question.

Partially recapitulating the foregoing results, we see in the case where the Absolute is not a point-pair, that the locus of a point such that the perpendiculars from it on the sides of the triangle have their feet in a line, is in general a cubic curve passing through the angles of the triangle: if, however, the condition

$$\frac{abc}{af^2} \cdot \frac{1}{bg^2} \cdot \frac{1}{ch^2} = \frac{1}{fgh} \quad (13)$$

be satisfied, that is, if the triangle be such that the angles thereof and of the reciprocal triangle lie in a conic (or, what is the same thing, if the sides touch a conic) then the cubic locus breaks up into the line

$$\frac{f}{a}x + \frac{g}{b}y + \frac{h}{c}z = 0, \quad (14)$$

which is the line through the points of intersection of the corresponding sides of the two triangles, and into the conic

$$\frac{A}{f}yz + \frac{B}{g}zx + \frac{C}{h}xy = 0, \quad (15)$$

which is the conic through the angles of the two triangles.

The question arises, given a conic (the Absolute) to construct a triangle such that its angles, and the angles of the reciprocal triangle in regard to the given conic, lie in a conic.

I suppose that two of the angles of the triangle are given, and I enquire into the locus of the remaining angle.

To fix the ideas, let A, B, C be the angles of the triangle, A', B', C' those of the reciprocal triangle; and let the angles A and B be given. We have to find the locus of the point C : I observe however, that the lines AA', BB', CC' meet in a point O , and I conduct the investigation in such manner as to obtain simultaneously the loci of the two points C and O .

The lines $C'B', C'A'$ are the polars of A, B respectively, let their equations be $x = 0$, and $y = 0$, and let the equation of the line AB be $z = 0$; this being so, the equation of the given conic will be of the form

$$(a, b, c, 0, 0, h \mid x, y, z)^2 = 0. \quad (16)$$

I take (α, β, γ) for the coordinates of C and (x, y, z) for those of O ; the coordinates of either of these points being of course deducible from those of the other.

Observing that the inverse coefficients are $(bc, ca, ab - h^2, 0, 0, -ch)$, we find

Coordinates of A are $(b, -h, 0)$,

B „ $(-h, a, 0)$.

The points A' and B' are then given as the intersections of AO with $C'A' (y = 0)$ and of BO with $C'B' (x = 0)$; we find

Moreover,

Coordinates of A' are $(h\alpha + b\beta, a\alpha + h\beta, h\gamma)$,

B' „ $(h\alpha + b\beta, a\alpha + h\beta, h\gamma)$,

Coordinates of C are $(0, 0, 1)$,

C' „ (α, β, γ) .

The six points A, B, C, A', B', C' are to lie in a conic; the equations of the lines CA, CB, AB are $hX + bY = 0, aX + hY = 0, Z = 0$, and hence the equation of a conic passing through the points C, A, B is

$$\frac{hX + bY}{Z} \cdot \frac{aX + hY}{Z} = 0. \quad (17)$$

Hence, making the conic pass through the remaining points A' , B' , C , we find

$$\frac{a\alpha(h\alpha + b\beta)}{L} + \frac{h(h\alpha + b\beta)}{M} + \frac{h\gamma}{N} = 0, \quad \frac{h(a\alpha + h\beta)}{L} + \frac{b(a\alpha + h\beta)}{M} + \frac{h\gamma}{N} = 0, \quad (18)$$

and eliminating the L , M , N , we find

$$\begin{vmatrix} 1 & a & 1 \\ 1 & h & 1 \\ 1 & h & 1 \\ h\alpha + b\beta & a\alpha + h\beta & h\gamma \\ a\alpha + h\beta & h\alpha + b\beta & z \end{vmatrix} = 0, \quad (19)$$

or developing and reducing, this is

$$\frac{ab - h^2}{\gamma} Y^2 = \frac{1}{a\alpha + h\beta} + \frac{1}{h\alpha + b\beta} - \frac{h\alpha\beta}{a\alpha + h\beta} - \frac{ab}{h\alpha + b\beta}, \quad (20)$$

that is

$$\frac{ab - h^2}{\gamma} Y^2 = (hb, -ab, ha \mid a\alpha + h\beta, h\alpha + b\beta)^2 = 0. \quad (21)$$

We have still to find the relation between (α, β, γ) and (x, y, z) ; this is obtained by the consideration that the line $A'B'$, through the two points A' , B' the coordinates of which are known in terms of (α, β, γ) , is the polar of the point C , the coordinates of which are (x, y, z) .

The equation of $A'B'$ is thus obtained in the two forms

$$(a\alpha + h\beta)Z + (h\alpha + b\beta)Y - \frac{(a\alpha + h\beta)(h\alpha + b\beta)}{ch\gamma} X = 0, \quad (22)$$

and

$$(ax + hy)X + (hx + by)Y + cz = 0, \quad (23)$$

and comparing these, we have

$$X : Y : Z = \alpha : \beta : \gamma = x : y : \frac{-(a\alpha + h\beta)(h\alpha + b\beta)}{ch\gamma}, \quad (24)$$

or what is the same thing

$$\alpha : \beta : \gamma = X : Y : \frac{-(a\alpha + h\beta)(h\alpha + b\beta)}{ch\gamma} = x : y : \frac{-(ax + hy)(hx + by)}{chz} \quad (25)$$

(where it is to be observed that the equation $\alpha : \beta = x : y$ is the verification of the theorem that the lines AA' , BB' , CC' meet in a point O).

We may now from the above found relation eliminate either the (α, β, γ) or the (x, y, z) ; first eliminating the (α, β, γ) , we find

$$\frac{ab - h^2}{\gamma} Y^2 = \frac{h}{a} \frac{1}{ax + hy} + \frac{1}{b} \frac{1}{hx + by} - \frac{hab}{(ax + hy)(hx + by)}, \quad (24)$$

where

$$\frac{Y}{Z} = \frac{(a\alpha + h\beta)(h\alpha + b\beta)}{ch\gamma^2} = \frac{(ax + hy)(hx + by)}{chz^2}, \quad (25)$$

or, completing the elimination,

$$\frac{ab - h^2}{\gamma} \frac{(ax + hy)^2(hx + by)^2}{chz^2} = (hb, -ab, ha \mid ax + hy, hx + by)^2 = 0, \quad (26)$$

which is a quartic curve having a node at each of the points

$$(z = 0, ax + hy = 0), \quad (z = 0, hx + by = 0), \quad (ax + hy = 0, hx + by = 0), \quad (27)$$

that is, at each of the points B, A, C .

The right-hand side of the foregoing equation is

$$= -(ab - h^2)(ha, ah, hh \mid x, y)^2 = -(ab - h^2)h(ax^2 + by^2 + \frac{2h}{ab}xy), \quad (28)$$

so that the equation may also be written

$$(ax + hy)^2(hx + by)^2 + ch^2z^2 \left(ax^2 + by^2 + \frac{2h}{ab}xy \right) = 0. \quad (29)$$

Secondly, to eliminate the (x, y, z) , we have

$$\frac{ab - h^2}{\gamma} Y^2 = \frac{1}{h} \frac{1}{a\alpha + h\beta} + \frac{1}{a} \frac{1}{h\alpha + b\beta}, \quad (30)$$

where

$$\frac{Y}{Z} = \frac{(a\alpha + h\beta)(h\alpha + b\beta)}{ch\gamma^2}, \quad (31)$$

or, completing the elimination,

$$(ab - h^2)ch\gamma^2 = (hb, -ab, ha \mid a\alpha + h\beta, h\alpha + b\beta)^2, \quad (32)$$

that is

$$= -(ab - h^2)h(a\alpha^2 + b\beta^2 + \frac{2h}{ab}\alpha\beta), \quad (33)$$

$$(a, b, c, 0, 0, h \mid \alpha, \beta, \gamma)^2 = 0. \quad (34)$$

Writing (x, y, z) in place of (α, β, γ) , the locus of the point is the conic

$$(a, b, c, 0, 0, h \mid x, y, z)^2 = 0, \quad (35)$$

which is a conic intersecting the Absolute $(a, b, c, 0, 0, h \mid x, y, z)^2 = 0$, at its intersections with the lines $x = 0, y = 0$, that is the lines CB and CA' .

In regard to this new conic, the coordinates of the pole of $C'B'$ ($x = 0$) are at once found to be $(-h, a, 0)$, that is, the pole of $C'B'$ is B ; and similarly the coordinates of the pole of $C'A'$ ($y = 0$) are $(b, -h, 0)$, that is, the pole of $C'A'$ is A .

We may consequently construct the conic the locus of O , viz. given the Absolute and the points A and B , we have $C'A'$ the polar of B , meeting the Absolute in two points (a_1, a_2) , and $C'B'$ the polar of A meeting the Absolute in the points (b_1, b_2) ; the lines $C'A'$ and $C'B'$ meet in C' .

This being so, the required conic passes through the points a_1, a_2, b_1, b_2 , the tangents at these points being Aa_1, Aa_2, Bb_1, Bb_2 respectively; eight conditions, five of which would be sufficient to determine the conic.

It is to be remarked that the lines $C'B', C'A'$ (which in regard to the Absolute are the polars of A, B respectively) are in regard to the required conic the polars of B, A respectively.

The conic the locus of O being known, the point may be taken at any point of this conic, and then we have A' as the intersection of $C'A'$ with AO , B' as the intersection of $C'B'$ with BO .

to the Absolute, the point so obtained being a point on the line CO .

To each position of O on the conic locus, there corresponds of course a position of C ; the locus of C is, as has been shown, a quartic curve having a node at each of the points C, A, B .

The foregoing conclusions apply of course to spherical figures; we see therefore that on the sphere the locus of a point such that the perpendiculars let fall on the sides of a given spherical triangle have their feet in a line (great circle), is a spherical cubic.

If, however, the spherical triangle is such that the angles thereof and the poles of the sides (or, what is the same thing, the angles of the polar triangle) lie on a spherical conic; then the cubic locus breaks up into a line (great circle), which is in fact the circle having for its pole the point of intersection of the perpendiculars from the angles of the triangle on the opposite sides respectively, and into the before-mentioned spherical conic.

395. Investigations in connexion with Casey's Equation.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. XII (1873), pp. .]

395.

The equation of Casey relates to the condition for four circles to touch a fifth circle. If $\rho_1, \rho_2, \rho_3, \rho_4$ are the radii of four circles touching a fifth circle of radius ρ , and t_{12}, t_{13} , etc. are the distances between their centres, then Casey's equation is

$$\begin{vmatrix} 0 & t_{12}^2 & t_{13}^2 & t_{14}^2 & 1 \\ t_{21}^2 & 0 & t_{23}^2 & t_{24}^2 & 1 \\ t_{31}^2 & t_{32}^2 & 0 & t_{34}^2 & 1 \\ t_{41}^2 & t_{42}^2 & t_{43}^2 & 0 & 1 \\ 1 & 1 & 1 & 1 & 0 \end{vmatrix} = 0. \quad (1)$$

If however the circles all touch the fifth circle externally, the equation takes a simpler form. Writing $t_{ij}^2 = (\rho_i + \rho_j)^2$ for external contact, or $t_{ij}^2 = (\rho_i - \rho_j)^2$ for internal contact, the determinant condition simplifies.

The proof depended on the properties of the circles of inversion. By inverting the system with respect to a circle whose centre is at one of the points of contact, the circles become parallel lines, and the condition is evident.

equation of

The equation may be written in the form

$$\begin{vmatrix} \rho_1 & \rho_2 & \rho_3 & \rho_4 \\ 1 & 1 & 1 & 1 \\ t_{23}^2 & t_{13}^2 & t_{12}^2 & \dots \\ \dots & \dots & \dots & \dots \end{vmatrix} = 0. \quad (2)$$

and substituting the values of t_{ij}^2 in terms of the radii, we obtain a relation which must be satisfied.

The condition for four circles to touch a fifth circle may also be expressed in terms of the curvatures. If the circles have curvatures $\kappa_1, \kappa_2, \kappa_3, \kappa_4$ (where $\kappa = 1/\rho$), and the distances between centres are t_{ij} , then the condition is that a certain determinant involving these quantities shall vanish.

The general theory of the contacts of circles may be applied to the determination of the circles which touch three given circles. The problem of Apollonius, to find a circle touching three given circles, has eight solutions, corresponding to the different combinations of external and internal contact.

The investigation of Casey's equation leads to many interesting results in the geometry of circles. The condition may be generalized to the case of spheres in three dimensions, and to higher dimensions.

The relation between the radii and the distances may be expressed in a symmetric form. Writing $u_{ij} = t_{ij}^2 / \rho_i \rho_j$, the condition becomes a relation between the quantities u_{ij} and the ratios of the radii.

The equation of Casey is a fundamental result in the theory of the contacts of circles. It has been applied to the study of the configurations of circles in a plane, and to the investigation of the properties of systems of circles.

The proof of Casey's equation may be obtained by considering the inversion of the system of circles. If we invert with respect to a circle whose centre is at the point of contact of two of the circles, these circles become parallel lines, and the condition for the contact of the remaining circles with the inverted circle is easily established.

396. On a certain Envelope depending on a Triangle inscribed in a Circle.

[From the *Proceedings of the London Mathematical Society*, vol. IV (1871), pp. 122–127.]

396.

Let a triangle be inscribed in a circle, and consider the envelope of a line connected with the triangle by a given relation.

We may take the circle to be $x^2 + y^2 = 1$, and the vertices of the triangle to be points on this circle.

and I proceed to find the equation of the envelope.

Let the vertices of the triangle be represented parametrically by the angles α, β, γ , so that the coordinates are $(\cos \alpha, \sin \alpha), (\cos \beta, \sin \beta), (\cos \gamma, \sin \gamma)$.

The equation of the line joining the points with parameters β and γ is

$$x \cos \frac{1}{2}(\beta + \gamma) + y \sin \frac{1}{2}(\beta + \gamma) = \cos \frac{1}{2}(\beta - \gamma). \quad (1)$$

We find the equation by eliminating the parameters between the equations of the line and the condition of envelopment.

Let the line be $\lambda x + \mu y + \nu = 0$, and let the condition of envelopment be that the line is connected with the triangle by a given relation, e.g. that it passes through a fixed point, or that it touches a given conic.

where the coefficients of the line equation are functions of the parameters determining the triangle.

The envelope of the line is obtained by eliminating the parameters between the equation of the line and the condition that the line touches the envelope. This leads to a curve of higher class.

I find the envelope to be a curve of the sixth class, having cusps and nodes depending on the particular relation assumed.

The equation of the envelope may be obtained by eliminating the parameters between the equations

$$\lambda \cos \alpha + \mu \sin \alpha + \nu = 0, \quad \lambda \cos \beta + \mu \sin \beta + \nu = 0, \quad \lambda \cos \gamma + \mu \sin \gamma + \nu = 0, \quad (2)$$

and the condition of envelopment.

I write the equation in the form

$$(a, b, c, d, e, f \mid x, y, z)^6 = 0, \tag{3}$$

where the coefficients are functions of the parameters determining the relation between the line and the triangle.

The class of the envelope is six, and the curve has in general nine cusps and four nodes. The particular case where the envelope degenerates into simpler curves is of special interest.